

Deriving the Multi-Dose Concentration Formula

From a First-Order Differential Equation via a Geometric Series
and Extending It to Multiple Daily Doses, Continuous Infusion, and Cessation

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Scope. This draft develops mathematical concentration-time models. It is not a dosing guide and does not prescribe, recommend, or evaluate treatment for any individual. Numerical drug examples are used only to illustrate the consequences of substituting reported population parameters into the formulas.

Research provenance. This paper grows out of more than 20 Mathematica notebooks created by the author. An early manuscript draft was partially assembled from the author's notebook material using Anthropic's Claude Fable 5. Subsequent manuscript development used OpenAI ChatGPT as an interactive mathematical and document-production interface for algebra checking, literature retrieval and comparison, plotting, organization, and LaTeX preparation. The research questions, motivating ideas, model directions, and judgments presented as the author's are the author's; ChatGPT is not treated as a source of the project's original research ideas or as an authority for mathematical correctness.

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Abstract

Beginning with the first-order elimination law, this paper develops a half-life-centered, base-two description of single and repeated drug concentration profiles. Equal-time offsets of exponential residual tails form a geometric series, giving finite-dose and steady-state peak/trough formulas. The framework develops D doses per day, practical elimination, the $D \rightarrow \infty$ continuous-infusion limit, cessation from an arbitrary concentration, and the passage-time function $T(C_a \rightarrow C_b)$, before addressing seven broader mathematical questions. For a general scalar autonomous decay law $C' = -f(C)$, a passage coordinate converts the dynamics into unit-speed translation, so passage time is a canonical coordinate rather than a peculiarity of exponential decay. Scale invariance of all passage times is proved to characterize first-order elimination among positive continuous autonomous laws. The resulting log-ratio metric and flow-semigroup formulation make first-order elimination conjugate to translation. Repeated arbitrary input kernels reduce to geometric residual tails after finite input ends, while denser fixed-total-amount dosing converges under natural kernel assumptions to continuous input. Multi-compartment cessation is recast as a vector-state problem, and an error-adjusted practical-elimination criterion is given for approximately first-order empirical curves. Drug-by-drug empirical validation remains for later work.

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1 Motivation and mathematical viewpoint

This project grew in part from a long-standing interest in understanding the mathematics of medications taken on a regular schedule. The mathematical question is deliberately broader than any particular medication: starting from a small set of dynamical assumptions, what can be derived exactly about accumulation, steady state, dose frequency, continuous input, and the behavior of concentration after input stops?

The governing principle throughout is a balance law. If $C(t)$ is concentration in a well-mixed effective compartment, then

$$V \frac{dC}{dt} = \text{rate in} - \text{rate out.} \quad (1)$$

The simplest post-input assumption is that the rate out is proportional to concentration. This produces the familiar exponential law, but the derivations below are arranged so that later replacements of the elimination law can be made transparently.

Notebook and AI provenance

The mathematical project predates this manuscript and is represented by more than 20 Mathematica notebooks created by the author. Those notebooks are the primary working record from which the repeated-dose construction developed. An early manuscript draft was partially assembled from the author's Mathematica notebook material using Anthropic's Claude Fable 5. OpenAI ChatGPT was subsequently used as an interactive interface to help inspect consequences of the author's questions, check algebra, retrieve and compare literature, produce explanatory plots, reorganize the exposition, and prepare LaTeX. The author does not use ChatGPT as the source of the project's original research ideas. Model output is treated as material to be checked, not as mathematical authority; authorship, interpretation, and final decisions remain with the author.

Prior art and originality boundary

Textbook material. Several ingredients used here are standard and are not claimed as original: solving the first-order equation $C' = -kC$; rewriting e^{-kt} as $2^{-t/h}$; linear superposition; finite and infinite geometric-series sums; the usual one-compartment bolus and constant-infusion equations; steady-state accumulation under linear kinetics; separation of variables; and the fact that first-order exponential decay becomes a straight line after taking a logarithm. **Parallel prior literature not used as the source of the author's ideas.** Retrospective literature checking found close precedents for several less elementary parts of the paper. Hajratwala derived generalized multiple-dose equations, including maxima and minima after n and infinitely many rapid intravenous doses under first-order elimination.[2] Norman developed one-compartment bolus and continuous-infusion equations explicitly from first principles.[1] Veng-Pedersen and Modi related steady-state peak-to-trough ratios to dosing intervals,[3] while Hughes, Glass, and Jacobs introduced the context-sensitive half-time as the time required for concentration to fall by 50% after an infusion is stopped,[4] and Bailey generalized this idea to “relevant decrement times” between clinically relevant concentration levels.[5] Veng-Pedersen also treated autonomous nonlinear elimination in the form $dC/dt = -q(C)$, [6] and later work derived analytical constant-infusion solutions when first-order and Michaelis–Menten elimination act simultaneously.[8] These papers were located during comparison and checking after the corresponding directions had arisen from the author's notebook work and questions; they are cited as parallel prior art, not as sources from which the author derived

those ideas. Author-developed synthesis not located in the targeted search. In the literature searches and accessible records checked for this manuscript, no source was located that presents the exact combination used here: a base-two, half-life-centered repeated-dose derivation with an explicit finite input window; the fixed-total-daily-amount $D \rightarrow$ infinity limiting bridge from increasingly frequent discrete dosing to continuous infusion in this normalization; the epsilon-practical-elimination criterion as a named theorem; and the particular synthesis that begins cessation at an arbitrary point of the periodic steady-state trajectory, writes the elapsed drop as $T(C_a \rightarrow C_b)$, and then connects that notation back to the peak-to-trough identity. This last category is therefore described only as not located in the surveyed literature, not as a proof of worldwide priority or mathematical novelty; some of its individual algebraic consequences are elementary once the model is specified.

2 Symbols and standing conventions

Symbol	Meaning
t	time
h	elimination half-life
x	duration of an input/absorption window
P	dosing period
D	equally spaced administrations per 24 hours, so $P = 24/D$
τ	time since the most recent dose finished its input window
A	total normalized amount administered per 24 hours
q	continuous concentration-input rate
C_a, C_b	two concentration levels with $C_a > C_b > 0$

The base-two convention is retained because it keeps the half-life visible. With h as half-life, one half-life multiplies concentration by $1/2$, two multiply it by $1/4$, and m half-lives multiply it by 2^{-m} .

3 First-order elimination and the half-life parameterization

Definition 3.1 (First-order elimination). *After input has stopped, suppose*

$$\frac{dC}{dt} = k_1 C, \quad k_1 < 0. \quad (2)$$

The separated solution is $C(t) = K e^{k_1 t}$. Requiring one half-life to halve the concentration gives

$$e^{k_1 h} = \frac{1}{2}, \quad k_1 = -\frac{\ln 2}{h}, \quad (3)$$

and therefore

$$\boxed{C(t) = C(0) 2^{-t/h}.} \quad (4)$$

4 Single-dose input: the original ramp and an exact zero-order input model

The original construction uses the normalized piecewise profile

$$C_{\text{ramp}}(t) = \begin{cases} \frac{t}{x}, & 0 < t \leq x, \\ 2^{-(t-x)/h}, & t > x, \end{cases} \quad (5)$$

which is continuous and normalized to equal one when the input window ends.

A useful mathematical comparison is an exact constant-rate (zero-order) input acting simultaneously with first-order elimination. During $0 < t \leq x$ let

$$\frac{dC}{dt} = q - kC, \quad k = \frac{\ln 2}{h}, \quad C(0) = 0. \quad (6)$$

Then

$$C(t) = \frac{q}{k} \left(1 - e^{-kt} \right). \quad (7)$$

Normalizing by requiring $C(x) = 1$ yields

$$C_{\text{zero}}(t) = \frac{1 - 2^{-t/h}}{1 - 2^{-x/h}}, \quad 0 < t \leq x. \quad (8)$$

After x , both normalized models share the same branch $2^{-(t-x)/h}$. Thus the distinction concerns the shape of the rise, not the post-input first-order decay.

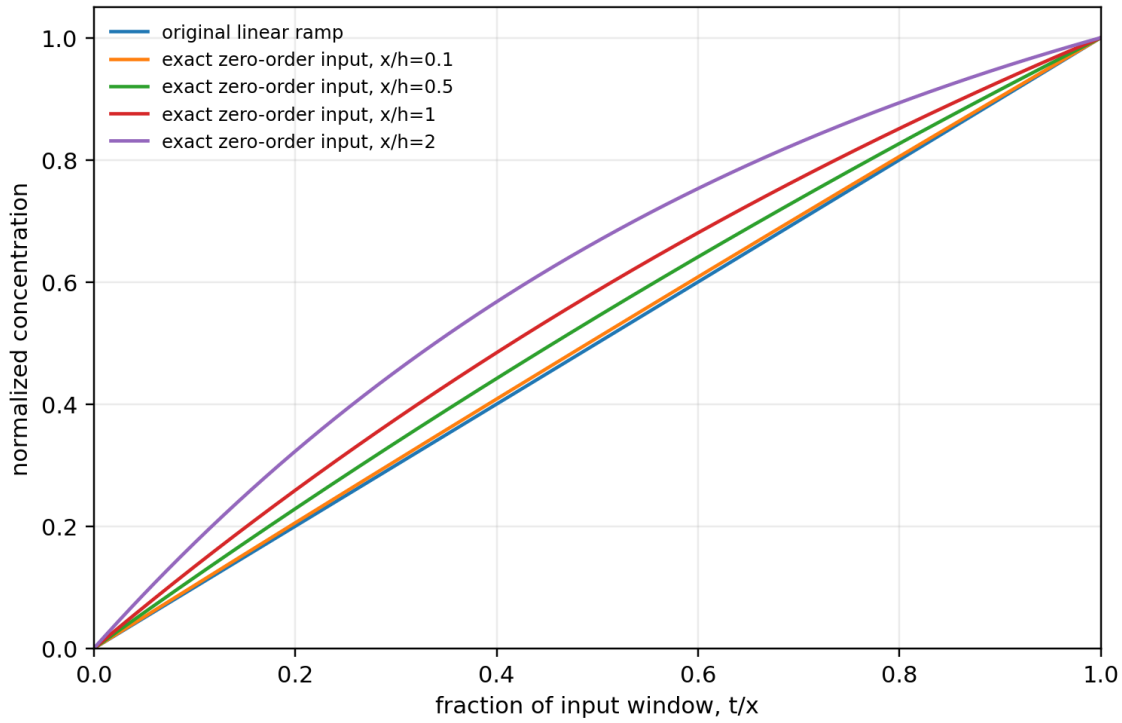


Figure 1: The original linear concentration ramp compared with the exact normalized solution produced by constant input plus simultaneous first-order elimination. When x/h is small, the two are close; as the input window becomes long relative to the half-life, simultaneous elimination bends the exact rise away from a straight line.

More generally, if $I(t)$ is an input rate already expressed in concentration-per-time units, the balance equation is

$$\frac{dC}{dt} = I(t) - kC. \quad (9)$$

Variation of constants gives

$$C(t) = C(0)2^{-t/h} + \int_0^t I(s) 2^{-(t-s)/h} ds. \quad (10)$$

This is the natural mathematical representation of how an administration is “packaged” in time: bolus, finite window, extended input, repeated input, and continuous infusion differ through $I(t)$.

5 Repeated dosing and the geometric series

Use the residual-tail construction. Doses are separated by period P . Closely related one-compartment multiple-dose equations for rapid intravenous injections under first-order elimination were derived by Hajratwala.[2] In the elimination phase of the current cycle, define τ as time since the most recent dose finished its input window. Then the contributions of the current and previous $n - 1$ doses are

$$C_n(\tau) = \sum_{i=0}^{n-1} 2^{-(\tau+iP)/h}. \quad (11)$$

Let

$$r = 2^{-P/h}, \quad 0 < r < 1. \quad (12)$$

Then

$$C_n(\tau) = 2^{-\tau/h} \frac{1 - r^n}{1 - r} = 2^{-\tau/h} \frac{1 - 2^{-nP/h}}{1 - 2^{-P/h}}. \quad (13)$$

Passing to the limit $n \rightarrow \infty$ with τ fixed gives

$$C_\infty(\tau) = \frac{2^{-\tau/h}}{1 - 2^{-P/h}}. \quad (14)$$

For the finite input window, $0 \leq \tau \leq P - x$. Hence

$$C_{\max,ss} = \frac{1}{1 - 2^{-P/h}} = \frac{2^{P/h}}{2^{P/h} - 1} \quad (15)$$

and

$$C_{\min,ss} = \frac{2^{-(P-x)/h}}{1 - 2^{-P/h}} = \frac{2^{x/h}}{2^{P/h} - 1}. \quad (16)$$

Their ratio is

$$\frac{C_{\max,ss}}{C_{\min,ss}} = 2^{(P-x)/h}. \quad (17)$$

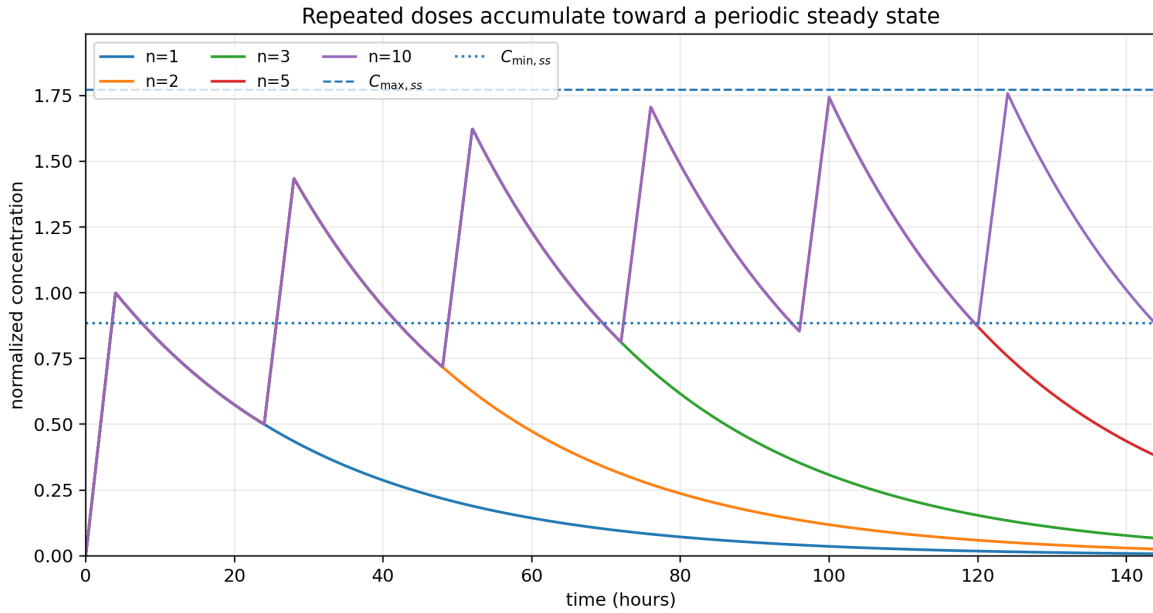


Figure 2: Repeated copies of the normalized profile accumulate toward a periodic steady state. The horizontal lines are the analytic steady-state peak and trough.

6 D equally spaced administrations per 24 hours

For $D > 0$ equally spaced administrations per day,

$$\boxed{P = \frac{24}{D}} \quad (18)$$

Substitution into the preceding formulas gives

$$C_{\infty,D}(\tau) = \frac{2^{-\tau/h}}{1 - 2^{-24/(Dh)}}. \quad (19)$$

Two different experimental questions must be separated.

6.1 Fixed amount per administration

If the amount attached to each administration is held constant while D increases, the total daily amount increases with D . The accumulation factor is

$$R_D = \frac{1}{1 - 2^{-24/(Dh)}}. \quad (20)$$

6.2 Fixed total daily amount

If instead the total normalized amount per 24 hours is fixed at A , then each of the D administrations has amplitude A/D . In the instantaneous-input idealization,

$$\boxed{C_{\max,ss}(D) = \frac{A/D}{1 - 2^{-24/(Dh)}}} \quad (21)$$

and

$$\boxed{C_{\min,ss}(D) = \frac{A/D \cdot 2^{-24/(Dh)}}{1 - 2^{-24/(Dh)}}} \quad (22)$$

For instantaneous boluses the peak-minus-trough jump is exactly A/D .

7 Practical elimination before the next administration

The first-order model never reaches literal zero at finite time. A tolerance is therefore needed whenever the phrase “eliminated before the next dose” is used mathematically.

Definition 7.1 (Practical elimination). *For $0 < \varepsilon < 1$, concentration is called ε -negligible after interval P when*

$$C(P) \leq \varepsilon C(0). \quad (23)$$

Theorem 7.2 (Practical elimination threshold). *Under first-order decay $C(t) = C(0)2^{-t/h}$, ε -negligibility by time P occurs if and only if*

$$\boxed{h \leq \frac{P}{\log_2(1/\varepsilon)}} \quad (24)$$

For D equally spaced administrations per 24 hours,

$$h \leq \frac{24}{D \log_2(1/\varepsilon)}. \quad (25)$$

Proof. The condition $C(P) \leq \varepsilon C(0)$ is equivalent to

$$2^{-P/h} \leq \varepsilon.$$

Taking $\log_2$ and rearranging gives (24). Substituting $P = 24/D$ gives the second statement. $\square$

For once-daily dosing ($D = 1$), the half-life boundaries corresponding to 5%, 1%, and 0.1% residual concentration after 24 hours are approximately 5.55 h, 3.61 h, and 2.41 h, respectively.

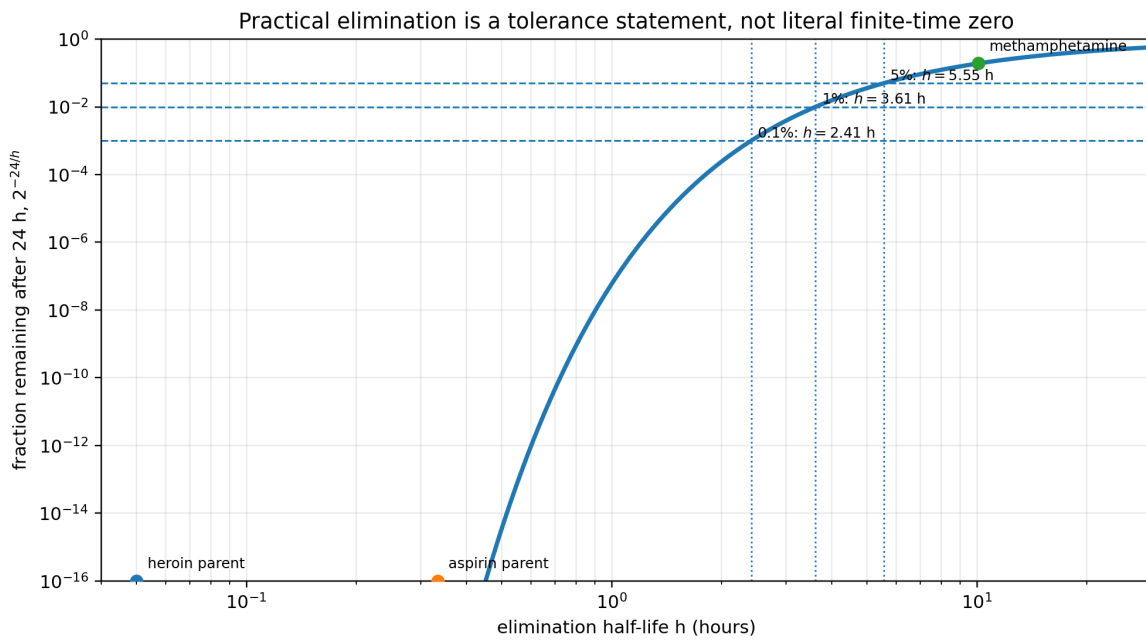


Figure 3: Residual fraction after 24 hours under first-order decay. The named drug points are only parameter illustrations for the parent compound: approximately 3 minutes for parenteral heroin in one human study, approximately 20 minutes for aspirin in a controlled extended-release study, and 10.1 hours as the mean methamphetamine half-life in a repeated-dosing human study. Metabolites, nonlinear elimination, formulation, and population variability are separate issues.

8 Continuous infusion as both an ODE solution and a discrete limit

8.1 Constant-input differential equation

Let q be a constant concentration-input rate. The standard one-compartment continuous-infusion equation can be developed directly from the compartment balance law; Norman gives an explicitly first-principles treatment of bolus and continuous intravenous input.[1] Then

$$\frac{dC}{dt} = q - kC, \quad k = \frac{\ln 2}{h}. \quad (26)$$

With $C(0) = C_0$,

$$C(t) = C_0 2^{-t/h} + \frac{qh}{\ln 2} \left(1 - 2^{-t/h}\right). \quad (27)$$

Starting from zero,

$$C(t) = C_{ss} \left(1 - 2^{-t/h}\right), \quad C_{ss} = \frac{qh}{\ln 2}. \quad (28)$$

Thus after 1, 2, 3, 4, and 5 half-lives, concentration is 50%, 75%, 87.5%, 93.75%, and 96.875% of steady state.

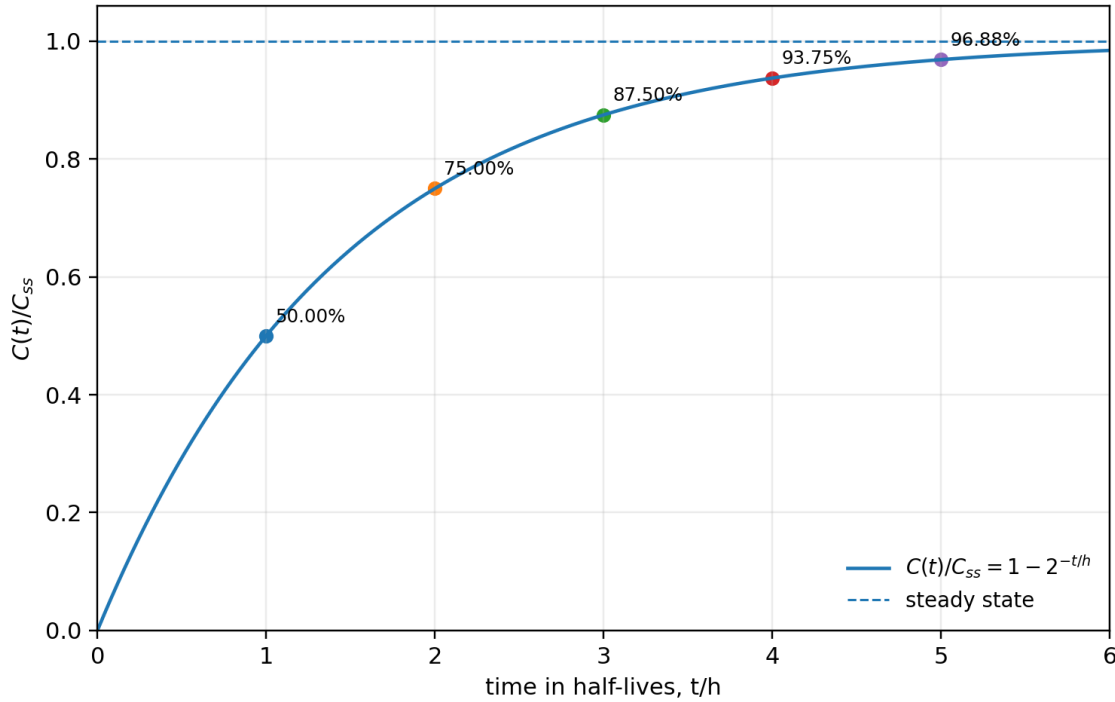


Figure 4: The constant-input solution approaches steady state by halving the remaining gap during each half-life.

Theorem 8.1 (Discrete-to-continuous limit at fixed total daily amount). *Let the total normalized amount per 24 hours be A . Split it into D equal instantaneous administrations, so each has amplitude A/D and period $P_D = 24/D$. Under first-order elimination,*

$$\lim_{D \rightarrow \infty} C_{\max,ss}(D) = \lim_{D \rightarrow \infty} C_{\min,ss}(D) = \frac{Ah}{24 \ln 2}. \quad (29)$$

This is exactly the continuous-input steady state obtained from (28) with $q = A/24$.

Proof. Write $z_D = 24 \ln 2 / (Dh)$. Since $1 - e^{-z_D} \sim z_D$ as $D \rightarrow \infty$,

$$\frac{A/D}{1 - 2^{-24/(Dh)}} = \frac{A/D}{1 - e^{-z_D}} \rightarrow \frac{Ah}{24 \ln 2}.$$

For the trough, multiply this expression by $2^{-24/(Dh)} \rightarrow 1$. The peak-trough gap is $A/D \rightarrow 0$. $\square$

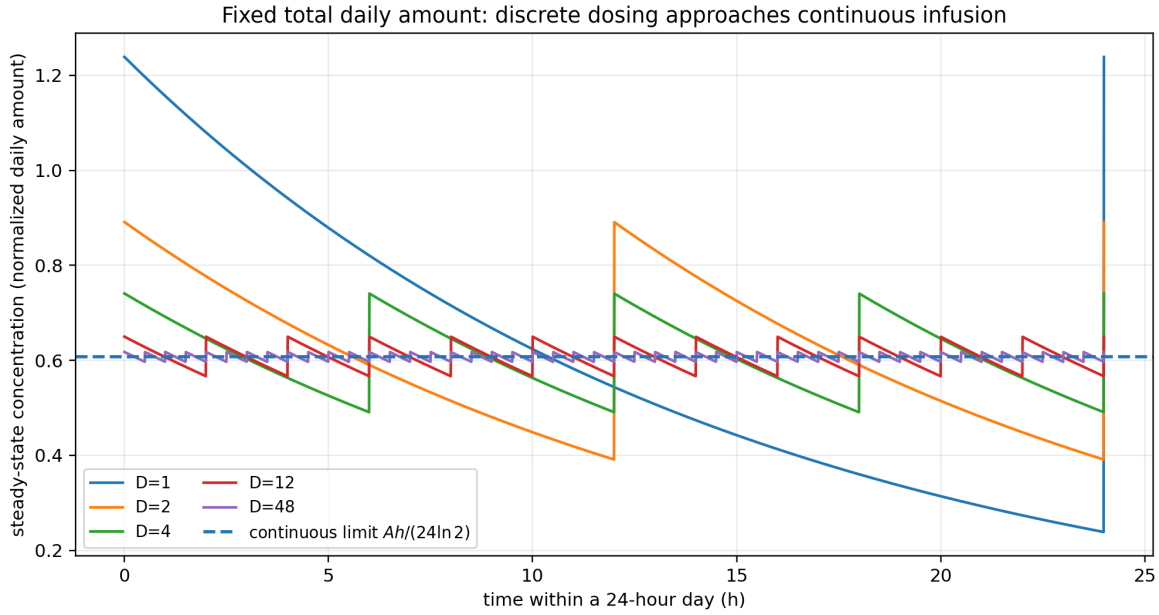


Figure 5: At fixed total daily amount, increasing D makes the steady-state oscillation smaller. Both discrete extrema converge to the constant-infusion level.

9 Cessation from an arbitrary point

The steady-state trough in the finite-window model is a limit in the number of continued doses; it is not the time-asymptotic value after dosing stops. If the next dose does not arrive, the concentration continues below the old trough.

Theorem 9.1 (Cessation from an arbitrary point). *Assume an arbitrary prior dosing history has produced concentration $C(T) = C_T > 0$ at cessation time T . If there is no further input and all remaining concentration obeys the same first-order law, then for $s \geq 0$,*

$$C(T + s) = C_T 2^{-s/h}. \quad (30)$$

The future trajectory depends on the prior dosing history only through the scalar state C_T .

Proof. After cessation the initial-value problem is $C' = -(\ln 2/h)C$ with $C(T) = C_T$. Solving yields (30). $\square$

If cessation occurs at a point τ_0 on the steady-state elimination branch (14), then

$$C(T + s) = \frac{2^{-(\tau_0+s)/h}}{1 - 2^{-P/h}}. \quad (31)$$

If cessation occurs exactly at the steady-state trough,

$$C_{\text{post-trough}}(s) = C_{\text{min,ss}} 2^{-s/h}. \quad (32)$$

The trough is therefore a distinguished point at which the next dose normally interrupts the decay, not a lower boundary imposed by the elimination equation.

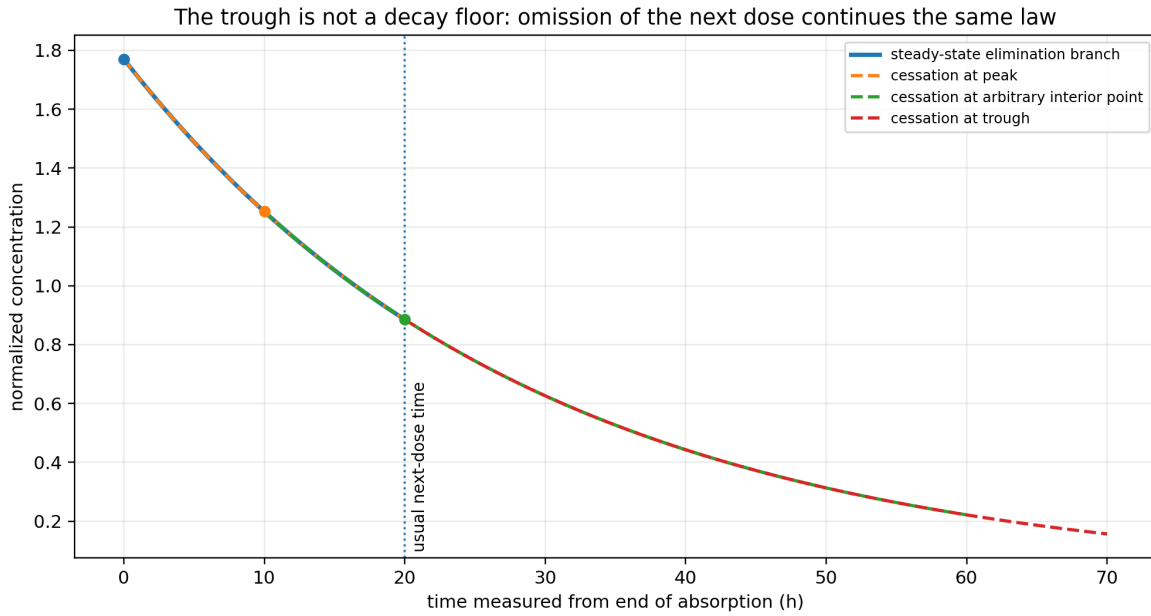


Figure 6: Cessation at the steady-state peak, at an arbitrary interior point, or at the trough. Once new input is removed, each trajectory continues by the same first-order law from its own starting state.

10 Concentration passage time

Definition 10.1 (Passage-time notation). *For $C_a > C_b > 0$, define*

$$\boxed{T(C_a \rightarrow C_b) := \text{elapsed time required for concentration to fall from } C_a \text{ to } C_b.} \quad (33)$$

The arrow means “from C_a to C_b ”; it is not a logical implication symbol in this notation.

Theorem 10.2 (First-order passage time). *Under first-order elimination,*

$$\boxed{T(C_a \rightarrow C_b) = h \log_2 \left(\frac{C_a}{C_b} \right).} \quad (34)$$

Proof. Set $C_b = C_a 2^{-T/h}$ and solve for T . □

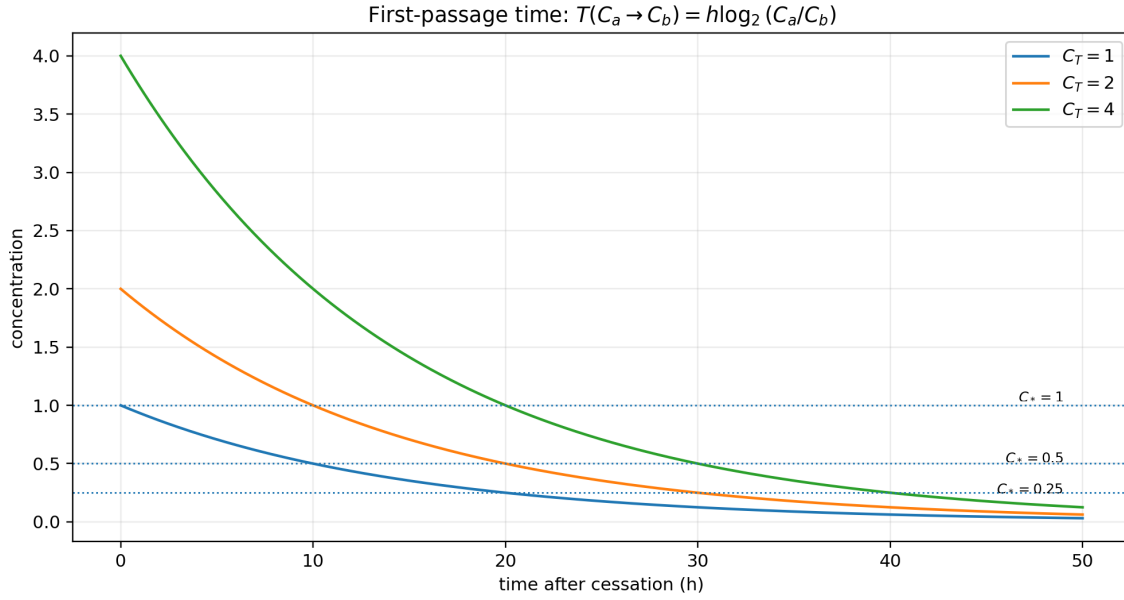


Figure 7: The time to reach a prescribed lower concentration depends on the ratio of starting to ending concentration and on the half-life.

Theorem 10.3 (Additivity and scale invariance). *If $C_0 > C_1 > C_2 > 0$, then*

$$\boxed{T(C_0 \rightarrow C_2) = T(C_0 \rightarrow C_1) + T(C_1 \rightarrow C_2).} \quad (35)$$

Moreover, for every $\lambda > 0$,

$$\boxed{T(\lambda C_a \rightarrow \lambda C_b) = T(C_a \rightarrow C_b).} \quad (36)$$

Proof. Using (34),

$$h \log_2 \frac{C_0}{C_2} = h \log_2 \frac{C_0}{C_1} + h \log_2 \frac{C_1}{C_2},$$

which proves additivity. Scale invariance follows because the common factor λ cancels inside the ratio. $\square$

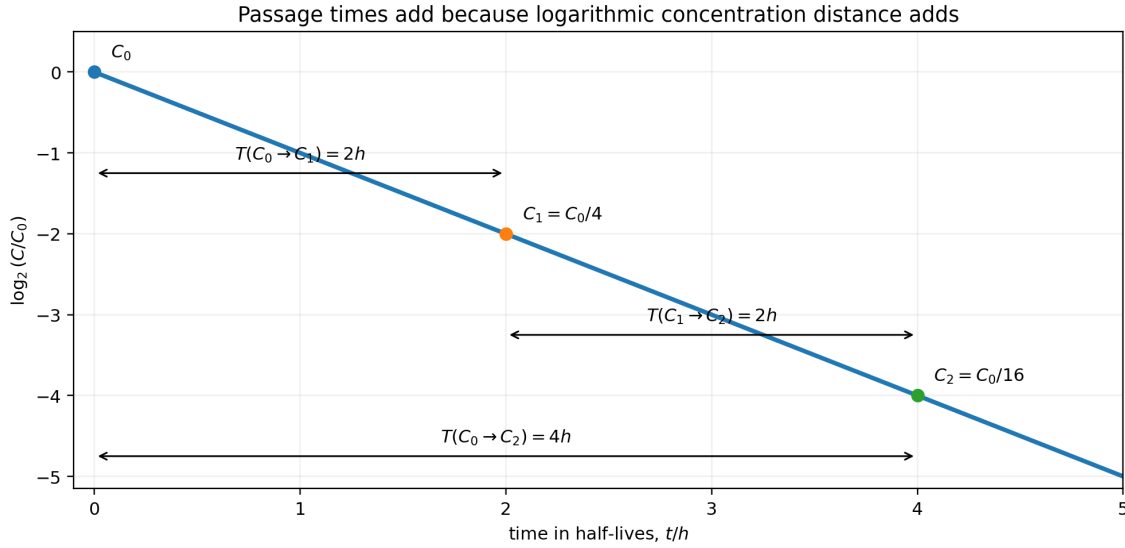


Figure 8: In logarithmic concentration coordinates, passage-time additivity is ordinary addition of adjacent line segments.

10.1 Peak-to-trough passage time

Theorem 10.4 (Peak-to-trough passage-time identity). For the steady-state elimination branch with input-window duration $x < P$,

$$\boxed{T(C_{\max,ss} \rightarrow C_{\min,ss}) = P - x.} \quad (37)$$

Equivalently, the steady-state fluctuation ratio and the elapsed peak-to-trough time encode the same information under first-order elimination.

Proof. Combining the fluctuation ratio (17) with Theorem 10.2 gives

$$\begin{aligned} T(C_{\max,ss} \rightarrow C_{\min,ss}) &= h \log_2 \left(\frac{C_{\max,ss}}{C_{\min,ss}} \right) \\ &= h \log_2 \left(2^{(P-x)/h} \right) \\ &= P - x. \end{aligned}$$

□

11 Logarithmic concentration coordinates

Theorem 11.1 (Logarithmic linearization of first-order elimination). *Define*

$$L(C) = h \log_2 C. \quad (38)$$

If $C(t) = C(0)2^{-t/h}$, then

$$\boxed{L(C(t)) = L(C(0)) - t.} \quad (39)$$

Hence exponential elimination in ordinary concentration is uniform linear motion with slope -1 in the coordinate $h \log_2 C$.

Proof.

$$\begin{aligned} L(C(t)) &= h \log_2 \left(C(0) 2^{-t/h} \right) \\ &= h \log_2 C(0) - t. \end{aligned}$$

□

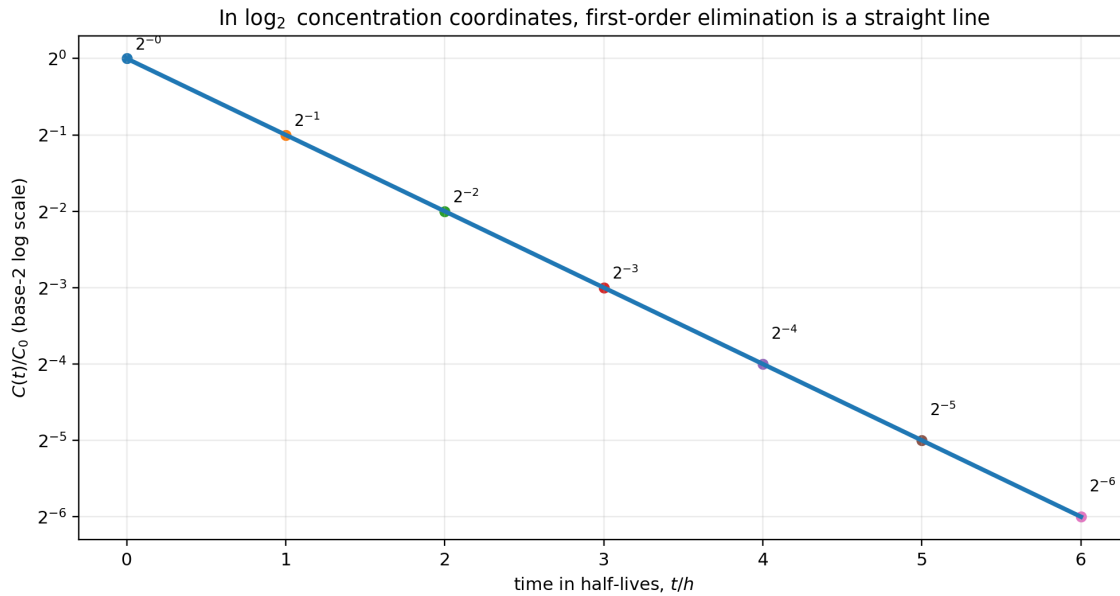


Figure 9: The exponential curve becomes a straight line when the vertical coordinate is logarithmic in base two. Each unit of t/h advances exactly one half-life.

This observation also clarifies cessation: the trough is simply one marked point on a straight logarithmic trajectory. Continued dosing supplies a new input before the line proceeds farther downward; cessation removes that interruption.

12 General autonomous elimination and first-passage time

The preceding passage-time structure extends beyond first-order elimination. Autonomous nonlinear pharmacokinetic formulations of the form $dC/dt = -q(C)$ have explicit precedent in the literature; Veng-Pedersen used this form in a model-independent steady-state analysis.[6] Suppose after input stops,

$$\frac{dC}{dt} = -f(C), \quad f(C) > 0. \quad (40)$$

Separation gives $dt = -dC/f(C)$.

Theorem 12.1 (General autonomous passage time). *For a solution of (40) that decreases from C_a to*

C_b , define $T(C_a \rightarrow C_b)$ as above. Then

$$T(C_a \rightarrow C_b) = \int_{C_b}^{C_a} \frac{dC}{f(C)}. \quad (41)$$

Whenever the integrals exist, passage times are additive.

The first-order model is recovered by $f(C) = kC$. Then

$$T(C_a \rightarrow C_b) = \frac{1}{k} \ln \frac{C_a}{C_b} = h \log_2 \frac{C_a}{C_b}.$$

A common nonlinear extension has simultaneous linear and saturable elimination,

$$\frac{dC}{dt} = -kC - \frac{v_{\max}C}{K_M + C}. \quad (42)$$

Here no arbitrary switching concentration is required: the governing equation remains the same while different terms dominate in different concentration regimes. At $C \ll K_M$, the Michaelis-Menten term is approximately $(v_{\max}/K_M)C$ and the tail again becomes approximately first order. Analytical constant-infusion solutions for this class of models have been studied in the pharmacokinetic literature.[7, 8]

13 Comparison with pharmacokinetic literature: derivation versus fitting

The central question is not whether pharmacokinetic formulas are “theoretical” or “empirical.” In practice there are at least three separate operations:

1. choose a structural balance model and its compartments/input laws;
2. derive or solve the mathematical concentration equations implied by that model;
3. estimate parameters, compare candidate models, or validate predictions using measured concentration data.

The sources below were included only where the method could be checked from an accessible abstract or article record; no claim is made about inaccessible textbook derivations.

Source	What the source says it does	Relation to the present derivation
Norman (1992)[1]	Develops one-compartment concentration equations “from first principles” for bolus, continuous IV input, and simulated IM input.	Supports the view that the standard linear infusion formula is an analytic consequence of a compartment balance law, not a curve-fitting identity.
Hajratwala (1976)[2]	Develops generalized multiple-dose equations for rapid IV injections under first-order elimination, including maxima/minima after n and infinitely many doses.	Closely parallels the geometric-series accumulation structure; the present paper emphasizes base two, an explicit input window, D doses/day, and cessation passage times.
Veng-Pedersen (1984)[6]	Develops model-independent steady-state predictions for autonomous nonlinear kinetics and explicitly writes the post-bolus law as $dC/dt = -q(C)$.	Provides direct prior literature context for the general autonomous decay form used in Section 12; the passage-time integral here is the elementary separation-of-variables consequence of that form.
Wu, Chen, Li (2021)[8]	Derives closed-form solutions for constant infusion with simultaneous first-order and Michaelis-Menten elimination.	Shows that analytic derivation continues beyond the linear model, although the functions required become more complicated.
Cook et al. (1992)[9]	Measures oral methamphetamine concentrations in human volunteers; reports that a one-compartment model with lag time fits the data best; reports mean half-life 10.1 h.	Example of empirical data being used to select/fit a structural model and estimate its parameters. The equation is not merely inferred from the fitted curve; the fitted model is chosen from a pharmacokinetic model class.
Pleasant et al. (1988)[10]	Collects serial tobramycin concentrations after IV infusion and fits data to a biexponential decay model with zero-order input.	Example in which measured data are explicitly fitted to a model in order to estimate pharmacokinetic quantities.

Table 1: A methodological distinction: analytic formulas and empirical parameter estimation often coexist in the same field.

For the standard one-compartment constant-infusion model, write the physical infusion rate as R_0 (amount/time) and effective volume as V . Then $q = R_0/V$ and

$$C_{ss} = \frac{R_0}{Vk} = \frac{R_0}{CL} = \frac{R_0 h}{V \ln 2}, \quad (43)$$

where $CL = Vk$. This is exactly equation (28) written in conventional clearance notation.

For rapid repeated IV boluses of amount B in volume V , the standard linear one-compartment steady-state extrema are

$$C_{\max,ss} = \frac{B/V}{1 - e^{-kP}}, \quad C_{\min,ss} = \frac{(B/V)e^{-kP}}{1 - e^{-kP}}. \quad (44)$$

Using $e^{-kP} = 2^{-P/h}$ gives the same base-two structure as equations (15)–(16) when the finite input-window offset is removed ($x = 0$).

14 Parameter illustrations, not clinical predictions

The following substitutions are included only to show how dramatically the dimensionless ratio P/h changes the mathematics. They do not model metabolites, tissue compartments, pharmacodynamic persistence, individual organ function, formulation effects, or nonlinear clearance.

For once-daily spacing, define the residual fraction

$$\rho_{24} = 2^{-24/h}. \quad (45)$$

A human heroin study reported a mean parent-heroin half-life of 3.0 ± 1.3 minutes after intravenous injection or infusion; an extended-release aspirin study describes aspirin as rapidly hydrolyzed with a half-life of about 20 minutes, while a separate intravenous aspirin study reported a terminal half-life of 15.4 ± 2.5 minutes; and a human methamphetamine repeated-dosing study reported a mean elimination half-life of 10.1 hours (range 6.4–15.1 h).[11, 12, 13, 9] These produce qualitatively different ρ_{24} values under the simple first-order parent-compound model: effectively negligible for the first two examples, but about 0.19 for the methamphetamine mean.

The word “parent” matters. Heroin is rapidly converted to 6-acetylmorphine and morphine, and aspirin has metabolites and pharmacodynamic effects that do not share the parent compound’s plasma half-life. The purpose of the examples is therefore mathematical classification by P/h , not a claim that the entire physiological effect is gone when the parent concentration is small.

15 Further structural questions and results

Seven structural questions arise naturally from the preceding development. This section addresses each one from the mathematics already developed, using outside literature only where the paper itself is insufficient. Each subsection states the question directly, gives the mathematical result or partial answer, and identifies any remaining empirical gap.

15.1 Question 1: arbitrary repeated input kernels

Question. How far can the geometric-series treatment be generalized from identical residual exponentials to arbitrary repeated input kernels through equation (10)?

Result. Let a single administration have a causal input kernel $g(t)$ and let the first-order elimination kernel be $e_k(t) = e^{-kt} \mathbf{1}_{t \geq 0}$. The concentration response to one administration is

$$r(t) = (g * e_k)(t). \quad (46)$$

For administrations at times $0, P, 2P, \dots, (N-1)P$ with amplitudes a_j , linearity gives

$$C_N(t) = \sum_{j=0}^{N-1} a_j r(t - jP). \quad (47)$$

Thus the exact geometric series is not a property of arbitrary kernels themselves; it appears whenever the translated residual response satisfies $r(t + P) = qr(t)$ on the region being summed. In particular, if g has finite support $[0, x]$, then after a dose has finished entering the compartment,

$$r(t) = B_g e^{-kt}, \quad t > x, \quad (48)$$

where

$$B_g = \int_0^x g(u) e^{ku} du. \quad (49)$$

Consequently all *completed* doses again form a geometric series with ratio $e^{-kP} = 2^{-P/h}$; only the finitely many doses whose input windows are still active require separate kernel terms.

Proposition 15.1 (Finite-window kernel decomposition). *For first-order elimination and an identical causal input kernel supported on $[0, x]$, the concentration at any time is the sum of (i) finitely many currently active input-window responses and (ii) a geometric series of completed-dose residuals with common ratio $2^{-P/h}$.*

Proof. For every completed dose, $t - jP > x$, so

$$r(t - jP) = e^{-k(t-jP)} \int_0^x g(u) e^{ku} du = B_g e^{-kt} e^{kjP}.$$

Successive completed-dose residuals therefore differ by the fixed factor e^{-kP} when indexed backward from the most recent completed dose. Doses for which $0 \leq t - jP \leq x$ remain inside the convolution window and form the finite active remainder. $\square$

Status. Substantial progress: the geometric-series structure survives much more generally than the original linear ramp, provided the input window is finite and elimination is first order. Infinite-support kernels require a different decomposition; sums of exponentials remain tractable term-by-term, but an arbitrary kernel need not produce a literal geometric series.

15.2 Question 2: passage time as a primary coordinate

Question. Can the passage-time function $T(C_a \rightarrow C_b)$ be used as the primary coordinate for comparing linear and nonlinear elimination laws?

Result. Yes, for any scalar autonomous decay law

$$\frac{dC}{dt} = -f(C), \quad f(C) > 0, \quad (50)$$

on an interval of positive concentrations. Choose a reference concentration C_* and define the *passage coordinate*

$$\Psi(C) = \int_{C_*}^C \frac{du}{f(u)}. \quad (51)$$

Then

$$\frac{d}{dt} \Psi(C(t)) = \Psi'(C) C' = \frac{1}{f(C)} [-f(C)] = -1. \quad (52)$$

Hence

$$\boxed{\Psi(C(t)) = \Psi(C(0)) - t.} \quad (53)$$

Moreover,

$$\boxed{T(C_a \rightarrow C_b) = \Psi(C_a) - \Psi(C_b)}. \quad (54)$$

Theorem 15.2 (Canonical passage coordinate). *Every one-dimensional autonomous strictly decreasing concentration law $C' = -f(C)$ with continuous $f > 0$ is transformed by $\Psi(C) = \int^C du/f(u)$ into uniform translation at unit speed. Passage time is therefore the coordinate difference in Ψ .*

For first-order elimination $f(C) = kC$, one obtains $\Psi(C) = k^{-1} \ln C + \text{const}$, which is the logarithmic coordinate already found in Section 11. Thus logarithmic linearization is the first-order member of a general passage-coordinate construction.

Status. Answered at the scalar autonomous level. This gives a common coordinate in which nonlinear and linear decay laws can be compared by the shape of the map $C \mapsto \Psi(C)$ even though every such law becomes translation after transformation.

15.3 Question 3: what uniquely characterizes first-order decay?

Question. Which properties characterize first-order decay uniquely among autonomous decay laws? In particular, additivity is general, but scale invariance is not.

Result. Scale invariance of *all* passage times does characterize first-order decay under mild regularity.

Theorem 15.3 (Scale invariance characterizes first-order elimination). *Let $f : (0, \infty) \rightarrow (0, \infty)$ be continuous and let*

$$T(a \rightarrow b) = \int_b^a \frac{du}{f(u)}, \quad a > b > 0.$$

Then the following are equivalent:

- (i) $T(\lambda a \rightarrow \lambda b) = T(a \rightarrow b)$ for all $a > b > 0$ and all $\lambda > 0$;
- (ii) $f(\lambda C) = \lambda f(C)$ for all $C, \lambda > 0$;
- (iii) $f(C) = kC$ for some constant $k > 0$.

Proof. Assume (i). Changing variables $u = \lambda v$ gives

$$\int_b^a \frac{\lambda dv}{f(\lambda v)} = \int_b^a \frac{dv}{f(v)}$$

for every interval (b, a) . Continuity implies equality of the integrands, so $\lambda/f(\lambda v) = 1/f(v)$ and hence $f(\lambda v) = \lambda f(v)$, proving (ii). Taking $v = 1$ yields $f(C) = Cf(1) = kC$, proving (iii). Conversely, if $f(C) = kC$, then $T(a \rightarrow b) = k^{-1} \ln(a/b)$, so the common factor λ cancels and (i) follows. $\square$

Status. Answered under continuity and positivity: passage-time additivity is generic, whereas scale invariance across every concentration rescaling singles out the first-order law.

15.4 Question 4: distance and semigroup structure

Question. Can a “distance” or semigroup structure on concentration states formalize the observation that first-order elimination becomes translation in logarithmic coordinates?

Result. Yes. For any scalar autonomous law, the canonical passage coordinate defines

$$d_f(C_1, C_2) = |\Psi(C_1) - \Psi(C_2)|. \quad (55)$$

Because Ψ is strictly increasing, d_f is a metric on the concentration interval. In first-order elimination,

$$\boxed{d_h(C_1, C_2) = h \left| \log_2 \frac{C_1}{C_2} \right|}. \quad (56)$$

The elimination flow is

$$\Phi_t(C) = C2^{-t/h}, \quad t \geq 0, \quad (57)$$

and satisfies the semigroup law

$$\Phi_{t+s} = \Phi_t \circ \Phi_s. \quad (58)$$

Under $L(C) = h \log_2 C$,

$$L(\Phi_t(C)) = L(C) - t, \quad (59)$$

so the multiplicative concentration semigroup is conjugate to the additive translation semigroup on the real line.

Theorem 15.4 (Log-ratio metric and translation semigroup). *First-order elimination on $(0, \infty)$ is an isometric translation flow in the metric $d_h(C_1, C_2) = h |\log_2(C_1/C_2)|$. Multiplying both concentrations by the same positive factor leaves this distance unchanged.*

Status. Answered. The “distance” intuition can be made literal, and the semigroup statement formalizes the memoryless continuation of the trajectory after cessation.

15.5 Question 5: metabolites and multi-compartment states

Question. How should active metabolites and multi-compartment states alter the statement that the future depends only on the scalar $C(T)$? In a higher-dimensional model the cessation state is a vector rather than one number.

Result. The scalar statement generalizes to a state-vector statement. After cessation, let

$$\dot{\mathbf{z}} = F(\mathbf{z}), \quad \mathbf{z}(T) = \mathbf{z}_T, \quad (60)$$

where components may represent central and peripheral compartments, parent compound, metabolites, or other kinetically relevant stores. Under the usual uniqueness assumptions on F , the future trajectory depends on the past only through the full state vector $\mathbf{z}_T$, not through the entire earlier history. For a linear compartment model,

$$\boxed{\mathbf{z}(T + s) = e^{As} \mathbf{z}_T}. \quad (61)$$

If measured plasma concentration is $C = c^\top \mathbf{z}$, two different vectors can satisfy the same scalar value $c^\top \mathbf{z}_T$ yet have different future concentrations because hidden compartments or metabolites later feed the measured compartment.

Proposition 15.5 (Vector-state cessation principle). *In a deterministic multi-compartment autonomous model with unique solutions, the complete post-cessation future is determined by the state vector at cessation. A single measured concentration $C(T)$ is sufficient only when it uniquely determines the kinetically relevant state on the model’s admissible state set.*

Status. Answered structurally. The one-compartment “history collapses to one number” principle survives as “history collapses to the Markov state vector.” What generally fails is the sufficiency of one plasma concentration.

15.6 Question 6: the discrete-to-continuous limit with overlapping input windows

Question. Under what assumptions does the discrete-to-continuous limit remain valid for non-instantaneous or overlapping input windows?

Result. Let g be a nonnegative causal input kernel normalized by

$$\int_0^\infty g(u) du = 1. \quad (62)$$

Let $P = 24/D$ and $q = A/24$, so each administration has amount $A/D = qP$. With dosing beginning at time 0, define

$$I_P(t) = qP \sum_{j=0}^{\lfloor t/P \rfloor} g(t - jP). \quad (63)$$

As $P \rightarrow 0$, this is a Riemann sum and, for continuous integrable g ,

$$\boxed{I_P(t) \longrightarrow q \int_0^t g(u) du.} \quad (64)$$

Thus the dense-dose limit is exactly constant input q only after the input kernel has been fully populated. If g is supported on $[0, x]$, then for every $t \geq x$,

$$I_P(t) \longrightarrow q. \quad (65)$$

If instead the schedule is idealized as extending indefinitely into the past, the corresponding two-sided Riemann sum converges directly to q at every observation time away from boundary effects. Because the first-order concentration solution is a continuous convolution operator on the input, convergence of the inputs in a suitable L^1 or uniform sense implies convergence of the concentration trajectories on finite time intervals.

Theorem 15.6 (Dense repeated-kernel limit). *For a normalized continuous integrable causal kernel g , fixed total daily amount A , and spacing $P = 24/D$, the repeated input $qP \sum_j g(t - jP)$ converges to the cumulative-input profile $q \int_0^t g(u) du$ when dosing starts at 0. If g has finite support $[0, x]$, this limit equals the constant rate $q = A/24$ for $t \geq x$; hence the continuous-infusion limit survives overlapping windows after a startup boundary layer of duration at most x .*

Status. Largely answered for identical normalized kernels under first-order linear elimination. The important correction is that a fixed-width non-instantaneous kernel produces a startup boundary layer; instantaneous continuous input is recovered immediately only when that boundary effect is removed or the kernel width shrinks appropriately with P .

15.7 Question 7: when is the practical-elimination theorem empirically adequate?

Question. Which empirical drug concentration curves are sufficiently close to the first-order model that the practical-elimination theorem is useful without nonlinear correction?

Result. No drug-by-drug empirical classification was completed in this analysis . A mathematical diagnostic can nevertheless be stated. On an elimination interval define the local apparent

half-life by

$$h_{\text{app}}(t) = - \left(\frac{d}{dt} \log_2 C(t) \right)^{-1} \quad (66)$$

whenever the derivative is negative. Exact first-order elimination is equivalent to constant h_{app} . Empirically, semilog linearity can therefore be checked by the variation of h_{app} or by residuals from a linear fit to $\log C$ versus time.

A useful robust version of the practical-elimination theorem follows if the observed trajectory satisfies

$$C(t) = C_0 2^{-t/h} e^{\eta(t)}, \quad |\eta(t)| \leq \delta \quad (67)$$

on the interval of interest. Then

$$\frac{C(P)}{C_0} \leq e^{\delta} 2^{-P/h}. \quad (68)$$

Therefore the sufficient condition for practical elimination at tolerance ε becomes

$$h \leq \frac{P}{\log_2(1/\varepsilon) + \delta/\ln 2}. \quad (69)$$

Proposition 15.7 (Error-adjusted practical-elimination criterion). *If the multiplicative departure from a first-order trajectory is bounded by $e^{\pm\delta}$ over $[0, P]$, then the displayed inequality is sufficient to guarantee $C(P) \leq \varepsilon C_0$.*

Status. Mathematical progress, but no empirical answer yet to “which drugs.” A defensible drug-level classification requires concentration-time datasets, a specified elimination interval, and an explicit acceptable deviation δ rather than reliance on a catalog half-life alone.

15.8 Remaining open problems

These seven questions are not equally resolved. Questions 2-5 admit clean structural answers at the level of autonomous ODEs; Questions 1 and 6 have broad answers under finite-window or normalized-kernel assumptions; Question 7 remains empirically open even though a quantitative adequacy criterion is now available. Further development should test the scale-invariance characterization and dense-kernel limit against prior mathematical literature and should apply the empirical criterion only to datasets for which the elimination phase is directly observable.

16 Conclusion

The repeated-dose derivation is built on one simple observation: equal-time offsets of first-order residual tails form a geometric series. From that starting point, the framework develops dose frequency, continuous infusion, practical elimination, cessation, passage time, and logarithmic coordinates, then places several of those ideas in a broader structure. Every scalar autonomous monotone decay law possesses a canonical passage coordinate in which decay is unit-speed translation; first-order elimination is distinguished because passage times are invariant under a common rescaling of concentration. This scale invariance forces $f(C) = kC$. The associated log-ratio metric makes the first-order elimination semigroup literally a translation flow. Finite-window input kernels preserve the geometric series for completed residuals, and dense fixed-total-amount dosing converges to continuous

input after a kernel-dependent startup boundary layer. In higher-dimensional models, cessation remains memoryless only with respect to the full state vector rather than one measured concentration. Finally, approximate semilog linearity yields an error-adjusted form of the practical-elimination theorem, although drug-by-drug empirical adequacy remains an open validation problem.

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